

IBRAHIM CHEHAB

MECHANICAL ENGINEERING STUDENT

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ABOUT ME

Motivated Mechanical Engineering student dedicated to achieving meaningful outcomes in vehicle engineering systems design, development, and testing. Enthusiastic about automotive culture and embracing dynamic challenges through hands-on mechanical experience. Proven expertise in 3D CAD modeling, technical problem-solving, and effective collaboration on engineering project teams. Proficient in employing rapid prototyping tools and precise mathematical modeling to drive innovative product development.

EDUCATION

Bachelor of Science in Mechanical Engineering | Expected Graduation: May 2029

University of Michigan-Dearborn

- Standing: Junior
- Relevant Coursework: Engineering Computations, Fluid Mechanics, Thermodynamics, SolidWorks and Fusion

PROFESSIONAL & PROJECT EXPERIENCE

Formula SAE Aerodynamics Team Member | University of Michigan-Dearborn | Sept 2025 – Present

- Designed and optimized race vehicle aerodynamics to increase downforce, directly applying vehicle engineering systems design principles.
- Utilized computational fluid dynamics software and physical wind tunnel models to validate engineering concepts and improve overall vehicle performance.
- Collaborated within a fast-paced mechanical project team environment to meet strict manufacturing and testing deadlines.

Mechanical Design & Prototyping Projects | Academic & Independent | Aug 2025 – Present

- Created comprehensive 3D CAD models and technical engineering drawings for mechanical components including polygon axles, slotted spring pins, and bolt circles, using SolidWorks.
- Operated a Bambu Lab P1S 3D printer with an AMS system to rapidly prototype parts and fabricate custom mechanical adapters for hardware integration.
- Developed advanced MATLAB and Python scripts for precise mathematical calculations, symbolic differential equation solving, and 3D trajectory visualization.

Group Leader & Specials Teacher | Wayne Metro & Hype Athletics | Jun 2025 – Present

- Managed structured programs and coordinated group dynamics for large groups of students, demonstrating strong organizational and leadership skills.
- Maintained high accountability, adaptability, and reliable attendance, establishing the work ethic necessary to commit to a 40-hour work week.

TECHNICAL SKILLS

SolidWorks

3D Printing

MS Office Suite

MATLAB

Rapid Prototyping

Data Analysis

Python

Aerodynamic Testing

Mechanical Troubleshooting

Blender

Mathematical Modeling

ADDITIONAL INFORMATION

- **Languages:** English, Arabic.
- **Interests:** Automotive Culture, Formula 1 Technical Regulations, Race Strategy Analysis.
- **Leadership & Volunteer:** High School Varsity Golf Captain, National Honor Society (227+ hours of community service).